

Pramo Jewels

Technology Stack Decision Record

Document ID: PJ-TSDR-001
Version: 1.0
Status: Approved for Initial Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Document approved technology choices, rationale and acceptable alternatives
Audience	Engineering, Architects, DevOps, QA, AI Coding Agents
Version	1.0

Approved Technology Decisions

Area	Selected	Reason	Alternatives
Frontend	Next.js + React + TypeScript	SEO, SSR, scalability	Nuxt, Remix
Styling	Tailwind CSS	Fast, consistent design system	CSS Modules, SCSS
Animation	GSAP + Framer Motion	Premium motion & micro-interactions	Motion One
3D	Three.js + React Three Fiber	Interactive jewellery viewer	Babylon.js
Backend	Laravel 12	Rapid, secure, mature ecosystem	NestJS, Django
Database	MySQL	Reliable relational storage	PostgreSQL
Cache	Redis	Fast cache, queues, sessions	Memcached
Auth	Laravel Sanctum + OTP	Secure SPA authentication	JWT only
Storage	S3-compatible Object Storage	Scalable media storage	Local storage
Payments	Gateway with UPI/Card/COD	Indian market support	Other gateways
Source Control	Git + GitHub	Industry standard	GitLab
CI/CD	GitHub Actions	Integrated automation	GitLab CI
Container	Docker	Consistent environments	Podman

Decision Criteria

Selections prioritize security, maintainability, developer productivity, performance, SEO, ecosystem maturity and long-term scalability.

Version Policy

Use current stable LTS releases where possible. Major upgrades require compatibility testing and an Architecture Decision Record (ADR).

Technology Governance

Any proposal to replace a core technology must include impact analysis, migration plan, proof of concept and stakeholder approval.

Review Schedule

Review the stack every six months or before major platform expansions such as mobile apps, AI services or microservice extraction.